
Readable but Not Writable: A Causal Test of an AI-Authorship Direction in Language-Model Residual Streams

Ari Holtzman and NeuriCo

Abstract

Linear probes can distinguish human- from machine-written text in language-model activations, but a readable feature need not be a causal control variable. We test whether a dense residual-stream direction fitted to content-paired human and machine continuations also changes how artificial a model’s own writing appears. We estimate authorship directions in Qwen2.5-7B-Instruct and its pretrained base checkpoint using HAP-E, intervene during autoregressive decoding, and score held-out generations with a MAGE-trained lexical detector and an independently trained public RoBERTa detector. Authorship is almost perfectly readable: the validation-selected layer reaches 0.9997 held-out AUROC in both checkpoints. The write-side result is different. Across 16 instruct-model prompts, moving from strength -2 to $+2$ changes the MAGE detector’s $P(\text{AI})$ by $+0.008$ (95% cluster-bootstrap CI $[-0.077, 0.095]$) and the public detector’s score by -0.077 ($[-0.246, 0.057]$). Neither effect is significant, the detectors disagree in sign, and nuisance-orthogonal, random, shuffled-label, formality, and Assistant-axis controls reveal no specific effect. A 12-prompt base-model replication is likewise null. Word count, repetition, semantic similarity, and perplexity do not indicate endpoint quality collapse. These results establish a read/write dissociation within the tested design: authorship information is linearly accessible, but the validation-selected direction is not a reliable generative control for AI-sounding prose at the tested layer and doses.

1 Introduction

Machine-generated text leaves a strong statistical trace. Text detectors exploit that trace at the surface level, and recent work shows that a language model’s own hidden states can separate human from machine prose with simple linear probes Quaremba et al. [2026], Vishnyakov and Gaintseva [2026]. This evidence raises a sharper mechanistic question: if a model encodes an “AI-authorship” direction, does writing along that direction make its output sound more artificial?

The answer matters for both interpretability and detection. A causal direction would connect a detector’s latent signal to the mechanism that produces detectable prose. It could also expose detector vulnerabilities: a direction that can be subtracted without harming meaning would provide a direct route to detector evasion. A null result is informative for a different reason. It would show that a probe can summarize consequences of generation without identifying a variable that generation uses in the same form.

Existing results do not resolve this distinction. Residual-stream probes establish readout and transfer, but they do not intervene during free generation Quaremba et al. [2026], Vishnyakov and Gaintseva [2026]. Activation addition can control topics, sentiment, and other high-level properties Turner et al. [2023], Zou et al. [2023], and sparse-autoencoder features can steer stylistic cues associated with artificial text Kuznetsov et al. [2025]. Work on the Assistant persona and self-authorship establishes

causal axes for persona and attribution judgments Lu et al. [2026], Ackerman and Panickssery [2025]. None tests whether a dense human-machine separator controls independently perceived AI-like writing while preserving content and quality.

Our main question. We ask whether a direction trained to *read* authorship can also *write* it. We use the content-paired HAP-E corpus Reinhart et al. [2025], fit directions at five layers of QWEN2.5-7B-INSTRUCT and QWEN2.5-7B Qwen Team [2024], select a layer using validation readout only, and add the selected direction during held-out generation. We score outputs with two detectors that are independent of the activation data and compare the direction against nuisance-orthogonal, random, shuffled-label, formality, Assistant-axis, and prompt-only controls. Figure 1 summarizes this read-write design.

The result is asymmetric. The selected layer reaches 0.9997 held-out AUROC in both checkpoints, and every tested layer reaches 0.9997–1.000. Yet, for the instruct checkpoint, changing the intervention strength from -2 to $+2$ shifts a MAGE-trained detector by only $+0.008$ (95% CI $[-0.077, 0.095]$) and a public ROBERTA detector by -0.077 ($[-0.246, 0.057]$). Both intervals cross zero, the signs disagree, all Holm-adjusted detector p -values equal 1.0, and the base replication is also null. The result is not a universal causal null: with 16 prompts, the primary study has 80% power only for a paired standardized effect near $d_z = 0.75$.

Our contributions are:

- We formulate a direct read-write test for dense authorship directions, separating held-out linear decodability from generation-time causal control.
- We conduct a paired intervention study with signed doses, two independent detector outcomes, matched-norm controls, nuisance residualization, and explicit quality checks.
- We show near-perfect authorship readout in both base and instruction-tuned Qwen2.5-7B, while finding no detector-consistent causal effect at the validation-selected layer and tested doses.
- We identify a bounded negative result: linear accessibility is not sufficient evidence of write-side control, while small effects and other causal directions remain plausible.

Section 2 positions the study, section 3 gives the design, section 4 reports the evidence, and section 5 discusses its scope and implications.

2 Related Work

Representation-based detection. Quaremba et al. [2026] show that residual activations support sample-efficient linear machine-text detection and transfer across domains and generators. SV-DETECT similarly uses layerwise logistic directions as features for robust detection under distribution shift Vishnyakov and Gaintseva [2026]. Both results support a continuous and linearly accessible authorship signal, but neither adds the fitted separator during generation. Our study makes that intervention the primary test.

Detection datasets also shape what such a signal can mean. HC3 pairs human and ChatGPT answers across domains Guo et al. [2023]; MAGE collects broad in- and out-of-distribution machine-text data Li et al. [2024]; and RAID stresses detectors across generators, decoding settings, domains, and attacks Dugan et al. [2024]. These benchmarks show that detector performance can change sharply under distribution shift. We therefore train the lexical outcome detector outside HAP-E, validate both outcomes on MAGE and RAID, and require agreement before making a strong causal claim.

Activation steering and style. Contrastive activation addition and representation engineering show that residual-stream interventions can control high-level behavior at inference time Turner et al. [2023], Zou et al. [2023]. Human evaluation of style vectors further shows that moderate interventions can change perceived style while stronger ones can reduce quality Diallo et al. [2026]. Kuznetsov et al. [2025] identify sparse-autoencoder features related to formality, repetition, complexity, and tone in artificial-text detection and causally steer individual features. These findings motivate signed doses, quality measures, and style controls. Unlike feature-level steering, we test the dense supervised separator itself.

Persona and self-authorship. The Assistant Axis is a dominant, steerable persona direction that is present in base models and repositioned by post-training Lu et al. [2026]. A detector could label polished Assistant prose as machine-written, so we estimate same-model Assistant and formality axes

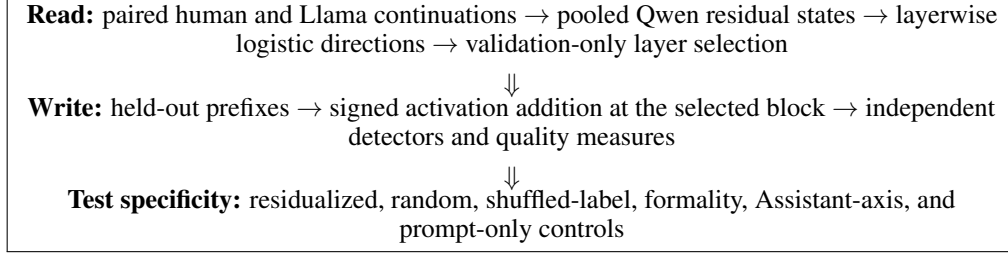


Figure 1: The read–write test. We fit authorship directions on content-paired continuations, select a layer using validation AUROC only, and intervene on separate held-out prompts. Outcome detectors never see the fitted Qwen activations. The same prompt is the paired unit across intervention strengths.

and remove their joint span from the authorship vector. Ackerman and Panickssery [2025] causally control whether Llama-3-8B-Instruct claims or perceives self-authorship. Their result is the closest mechanistic precedent, but changing a model’s answer to “did I write this?” is distinct from changing the surface properties judged by an independent observer.

Content-paired human and machine writing. Reinhart et al. [2025] introduce content-aligned corpora and show persistent grammatical and rhetorical differences between human and model continuations, with larger differences for instruction-tuned than base Llama variants. Their HAP-E structure lets us hold the prefix and document identity fixed. We use it both to estimate paired authorship directions and to compare base versus instruction-tuned Qwen checkpoints without treating topic as authorship.

3 Methodology

3.1 Question and estimands

Let $h_{i,y}^{(\ell)} \in \mathbb{R}^d$ be the attention-mask mean-pooled residual state at decoder layer ℓ for document i and continuation class $y \in \{0, 1\}$, where 0 denotes human and 1 denotes machine. We fit an L_2 -regularized logistic classifier on training documents,

$$P(y = 1 \mid h) = \sigma\left((\mathbf{w}^{(\ell)})^\top h + b^{(\ell)}\right), \quad (1)$$

and treat its normalized normal $\hat{\mathbf{w}}^{(\ell)} = \mathbf{w}^{(\ell)} / \|\mathbf{w}^{(\ell)}\|_2$ as the readout direction. To preserve the empirical scale of the paired contrast, we multiply this unit vector by the norm of the paired mean difference

$$\boldsymbol{\delta}^{(\ell)} = \frac{1}{n} \sum_i \left(h_{i,1}^{(\ell)} - h_{i,0}^{(\ell)} \right), \quad \mathbf{d}^{(\ell)} = \|\boldsymbol{\delta}^{(\ell)}\|_2 \hat{\mathbf{w}}^{(\ell)}. \quad (2)$$

Layer selection maximizes validation AUROC and never uses causal outcomes.

During cached autoregressive decoding, we intervene after the selected decoder block on each newly decoded token state h_t :

$$h'_t = h_t + \alpha \mathbf{d}, \quad (3)$$

where α is a signed dose. We leave prompt-prefill states and first-token logits unchanged. The primary causal estimand for outcome s is the paired prompt-average endpoint difference $\mathbb{E}_i[s_{i,+2} - s_{i,-2}]$. For the instruct five-dose sweep, we also estimate a linear slope within each prompt and average those slopes.

3.2 Data and splits

We use HAP-E’s shared-prefix structure Reinhart et al. [2025]. Human chunk 1 provides the prompt; human chunk 2 and the Llama-3-8B-Instruct continuation provide a content-paired human–machine contrast. We sample 400 unique document IDs across six genres and split them into 240 training, 80 validation, and 80 test IDs. The splits are document-disjoint, contain no missing text or duplicate IDs, and are fixed with seed 42.

Corpus lengths differ across generators: human chunk 2 averages 479 words, while model files average 426–579 words. We therefore truncate all readout inputs to the same 384-token budget and attention-mask mean-pool their states. Causal comparisons remain paired within the same prompt.

3.3 Models, layers, and nuisance directions

We study QWEN2.5-7B-INSTRUCT (revision a09a35458c702b33eeacc393d103063234e8bc28) and QWEN2.5-7B (revision d149729398750b98c0af14eb82c78cfe92750796) Qwen Team [2024]. For each checkpoint, we capture layers $\{5, 9, 14, 18, 22\}$ of 28 decoder layers and fit both the logistic normal and paired mean difference. Validation AUROC selects layer 5 for each checkpoint.

We estimate same-model Assistant and formality axes from eight real contrastive generation pairs per axis. Let $N = [d_{\text{asst}}, d_{\text{form}}]$ collect the two nuisance directions. We remove their joint span,

$$d_{\text{res}} = d - N(N^\top N)^{-1}N^\top d, \quad (4)$$

and renormalize d_{res} to the raw direction’s norm. We also measure pairwise cosine similarities among all three axes.

3.4 Interventions and baselines

For QWEN2.5-7B-INSTRUCT, we evaluate the raw authorship vector at $\alpha \in \{-2, -1, 0, +1, +2\}$ on 16 held-out test prefixes. We evaluate the following vector controls at $\alpha \in \{-2, +2\}$ on the same prompts: the fully residualized direction, an equal-norm isotropic random vector, a shuffled-label logistic vector, and direct formality and Assistant axes. The behavioral baseline prepends: “*Write in a natural, personal human style. Avoid polished assistant-like phrasing.*” For QWEN2.5-7B, we run a reduced replication using the raw direction at $\alpha \in \{-2, 0, +2\}$ on 12 held-out prefixes. All generations use greedy decoding and at most 80 new tokens.

3.5 Independent outcomes and quality measures

Our first outcome is a logistic detector trained on 24,000 balanced MAGE texts Li et al. [2024], with word and character TF-IDF features. Its training texts are independent of HAP-E and the Qwen activations. The second is the public `openai-community/roberta-base-openai-detector`, an independently pretrained and fine-tuned GPT-2-era detector OpenAI [2019]. Before causal interpretation, we validate both detectors on held-out MAGE and no-attack RAID data Dugan et al. [2024]; Appendix table 4 reports these checks.

We measure output quality and preservation with base-Qwen continuation perplexity, cosine similarity between all-MiniLM-L6-v2 embeddings of the steered and same-prompt unsteered continuations Reimers and Gurevych [2019], word count, unique-word ratio, and repeated 3-gram rate. These are proxies, not substitutes for blinded human judgments.

3.6 Statistical analysis and implementation

The prompt/document ID is the experimental unit. We form 95% intervals with 5,000 prompt-cluster bootstrap samples, use a 20,000-draw paired sign-flip permutation test, and report paired standardized effects $d_z = \bar{\Delta}/\text{SD}(\Delta)$. We apply Holm correction over detector/control comparisons and use a prespecified two-sided level of 0.05. Spearman correlation over dose means is descriptive.

We run bfloat16 Qwen inference with activation batch size 4 on one NVIDIA RTX A6000 (49,140 MiB), using Python 3.12.8, PyTorch 2.7.1+cu128, Transformers 5.18.0, scikit-learn 1.9.1, SciPy 1.18.1, pandas 3.0.6, NumPy 2.5.3, and statsmodels 0.15.0. Appendix section A provides exact commands and runtimes.

4 Results

4.1 Authorship is almost perfectly readable

The validation-selected layer is layer 5 for both checkpoints. It reaches 0.9997 held-out AUROC and 0.9813 accuracy in each case (table 1). Readout is not confined to the selected layer: all five tested

Table 1: Held-out authorship readout and direction geometry. Both checkpoints select layer 5 by validation AUROC. Bold marks the best value within each metric; ties are both bolded. Cosines use same-checkpoint directions.

Checkpoint	Layer	Val. AUROC	Test AUROC	Test acc.	cos(AI, Asst.)	cos(AI, Form.)	cos(Asst., Form.)
QWEN2.5-7B-INSTRUCT	5 / 28	0.9991	0.9997	0.9813	-0.021	+0.062	+0.827
QWEN2.5-7B	5 / 28	0.9992	0.9997	0.9813	-0.015	-0.046	+0.645

Table 2: Primary raw-direction endpoint effects ($\alpha = +2$ minus -2). Confidence intervals use 5,000 prompt-cluster bootstrap samples; p -values use 20,000 paired sign flips. No interval excludes zero, and all Holm-adjusted detector p -values equal 1.0.

Checkpoint	Outcome	n	Mean difference	95% CI	d_z	Perm. p
QWEN2.5-7B-INSTRUCT	MAGE $P(\text{AI})$	16	+0.008	$[-0.077, +0.095]$	+0.043	.864
QWEN2.5-7B-INSTRUCT	Public RoBERTA $P(\text{AI})$	16	-0.077	$[-0.246, +0.057]$	-0.241	.390
QWEN2.5-7B	MAGE $P(\text{AI})$	12	+0.032	$[-0.020, +0.088]$	+0.323	.294
QWEN2.5-7B	Public RoBERTA $P(\text{AI})$	12	+0.033	$[-0.061, +0.172]$	+0.147	.874

layers achieve 0.9997–1.000 test AUROC. Under this within-checkpoint design, instruction tuning does not measurably amplify separability. Because the directions are fit separately in each checkpoint, this comparison establishes representational availability, not vector identity across models.

The authorship direction is nearly orthogonal to the measured nuisance axes. In the instruct model, $\cos(\text{AI}, \text{Assistant}) = -0.021$ and $\cos(\text{AI}, \text{formality}) = 0.062$; the corresponding base-model values are -0.015 and -0.046 . By contrast, Assistant and formality overlap strongly (0.827 instruct; 0.645 base). The raw separator is therefore not simply either nuisance axis under these operationalizations.

4.2 Writing along the direction does not produce a consistent detector response

Table 2 reports the primary endpoint tests. On the 16 instruct prompts, moving from $\alpha = -2$ to $+2$ changes the MAGE detector’s $P(\text{AI})$ by +0.008 (95% CI $[-0.077, 0.095]$, $d_z = 0.043$, $p = .864$). The public detector moves in the opposite direction: -0.077 ($[-0.246, 0.057]$, $d_z = -0.241$, $p = .390$). The 12-prompt base replication yields small positive point estimates, but both intervals again cross zero. Every Holm-adjusted detector p -value is 1.0.

The full instruct sweep leads to the same conclusion (figure 2). Mean within-prompt slopes are $+0.0050 P(\text{AI})/\alpha$ ($[-0.0122, 0.0234]$) for the MAGE detector and -0.0186 ($[-0.0707, 0.0247]$) for the public detector. Descriptive correlations over the five dose means are $\rho = 0.40$ and $\rho = -0.50$, respectively, neither significant. Thus, the data show neither a robust dose response nor agreement between independent measures.

4.3 Residualization and matched controls do not reveal specificity

For the MAGE detector, endpoint differences are -0.027 ($[-0.147, 0.088]$) for the fully nuisance-orthogonal direction, $+0.008$ ($[-0.073, 0.103]$) for the random direction, $+0.008$ ($[-0.064, 0.096]$) for shuffled labels, $+0.020$ ($[-0.053, 0.097]$) for formality, and -0.007 ($[-0.106, 0.086]$) for the Assistant axis. Public-detector intervals also all cross zero (figure 3). Removing the measured nuisance span therefore does not uncover a hidden effect, and the raw vector has no demonstrated specificity over matched controls.

4.4 Endpoint quality does not explain the null

Across raw instruct endpoints, word count changes by +3.19, repeated 3-gram rate by -0.049 , semantic similarity to the baseline by $+0.002$, and base-Qwen perplexity by -6.58 (table 3). Every interval crosses zero. The base-model changes are also small: $+0.17$ words, -0.016 repetition, -0.015 similarity, and -0.04 perplexity. These measures do not show that output quality collapsed before an authorship effect could appear. They also do not prove preservation, because individual generations can change substantially and automated proxies miss important failures.

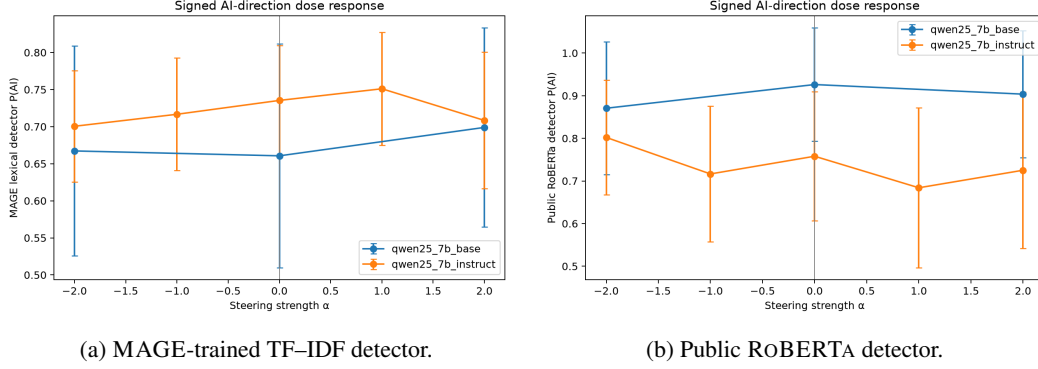


Figure 2: Signed dose response for the raw authorship direction. Points aggregate scores at each dose; uncertainty is computed over prompts. The instruct-model mean within-prompt slopes are $+0.0050$ $P(AI)/\alpha$ (95% CI $[-0.0122, 0.0234]$) for MAGE and -0.0186 $P(AI)/\alpha$ (95% CI $[-0.0707, 0.0247]$) for ROBERTA. Neither detector shows a robust monotonic response, and their slope signs disagree.

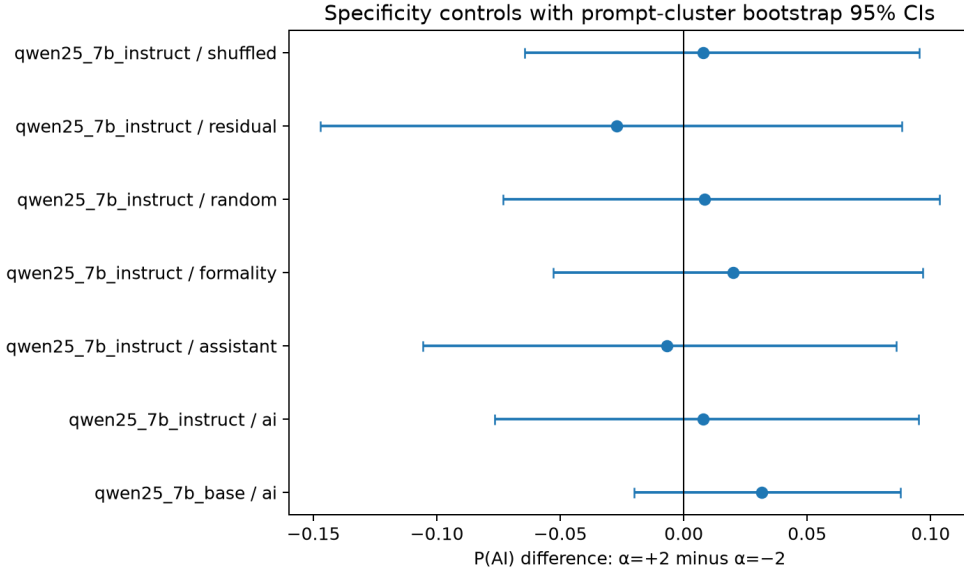


Figure 3: Endpoint effects ($\alpha = +2$ minus -2) for the instruct checkpoint. Error bars are 95% prompt-cluster bootstrap intervals. All detector intervals cross zero for the raw, nuisance-residualized, random, shuffled-label, formality, and Assistant directions. The raw authorship vector does not outperform equal-norm controls.

The corrected prompt-only “human style” baseline is not a successful alternative. Relative to unsteered generation, it changes MAGE $P(AI)$ by -0.022 ($[-0.133, 0.091]$) and public $P(AI)$ by $+0.060$ ($[-0.096, 0.213]$), again with opposite signs. It also decreases semantic similarity by 0.505 and increases repeated 3-grams by 0.212; several generations are repetitive or pathological, making mean perplexity extremely unstable. Appendix table 6 gives the complete comparison.

4.5 Large prompt-level shifts are heterogeneous

Aggregate null effects do not mean that every generation remains unchanged. For `blog_1119`, the endpoint shift is $+0.483$ for MAGE and $+0.158$ for the public detector. For `news_0877`, MAGE decreases by 0.396 while the public score changes by only -0.013 . For `acad_0130`, MAGE decreases by 0.237 while the public detector increases by 0.384. These cases show that individual

Table 3: Quality and preservation changes for the raw instruct direction ($\alpha = +2$ minus -2 , $n = 16$). No 95% prompt-cluster bootstrap interval excludes zero. Similarity is measured against the same prompt’s unsteered continuation.

Measure	Mean difference	95% CI
Word count	+3.19	$[-2.19, +8.63]$
Repeated 3-gram rate	−0.049	$[-0.119, +0.010]$
Semantic similarity	+0.002	$[-0.076, +0.078]$
Base-Qwen perplexity	−6.58	$[-19.94, +1.18]$

outputs can move substantially, but the direction and evaluator response are prompt-specific rather than a shared causal trend.

5 Discussion

What does the dissociation mean? The experiment separates two claims that are often treated as interchangeable. A probe can recover authorship nearly perfectly from a residual stream, yet adding its normal need not control the model’s output along the same semantic dimension. The fitted separator may aggregate downstream correlates of machine generation—regularity, lexical choices, or state geometry—without aligning with a write-side mechanism. Our evidence supports that interpretation within the tested setup, but it does not identify which correlates drive readout.

The geometry narrows one simple explanation. The authorship vector is nearly orthogonal to our same-model Assistant and formality axes, and removing their joint span does not produce a causal effect. Still, nuisance axes estimated from eight contrastive pairs are incomplete operationalizations. Near-zero cosine similarity does not rule out nonlinear overlap, distributed subspaces, token-dependent effects, or other stylistic variables.

The base-versus-instruct comparison also challenges a simple amplification story. Both checkpoints select layer 5 and reach the same 0.9997 held-out AUROC, while neither shows a reliable endpoint effect. Because we estimate vectors independently, we cannot conclude that post-training preserves the same geometric direction. The causal intervals are also too wide to resolve a subtle difference in leverage.

5.1 Limitations and threats to validity

Power. The causal experiment is small. With 16 paired prompts, 80% power requires approximately $d_z = 0.75$; with 12 prompts, it requires approximately $d_z = 0.89$. The study excludes a stable large effect more strongly than a small but meaningful one.

Layer and intervention choice. Validation AUROC selects early layer 5, but the most readable layer need not be the most causally effective. We avoid post-hoc layer shopping at the cost of limited causal localization. Equation 3 modifies only newly decoded token states after the first token; prompt-state interventions, multiple adjacent layers, projection removal, or activation capping may behave differently.

Direction data and evaluators. The machine half of the paired direction comes from Llama-3-8B-Instruct, not the target Qwen checkpoint. This choice strengthens cross-generator interpretation but may weaken write-side alignment. Both evaluators are automated: the lexical detector can react to surface artifacts, and the public ROBERTA detector was trained on older GPT-2-style text. Their sign disagreement is part of the result, but it also reduces sensitivity. A planned current-model blind judge could not run because the supplied API key had exhausted its daily limit; we did not replace missing ratings with simulated scores.

Quality and generality. Perplexity, sentence-embedding similarity, length, and repetition do not replace blinded human judgments of coherence and content. Results cover English HAP-E text, one 7B model family, one selected layer, and the tested dose range. The correct conclusion is “no detected effect here,” not “no causal AI-authorship direction exists.”

5.2 Broader implications

The methodological implication is direct: high probe accuracy should motivate a causal test, not stand in for one. Mechanistic claims are stronger when an intervention moves an outcome measured independently of the representation-fitting data and survives matched controls. Negative write-side evidence is therefore useful even when readout is nearly perfect.

A reliable authorship control could have dual uses. It might help analyze post-training style and detector failure modes, but it could also enable provenance evasion. This experiment does not provide such a method: neither signed activation addition nor explicit prompting reliably lowers both detector scores while preserving quality. Future work should nevertheless evaluate evasion risks alongside interpretability benefits.

6 Conclusion

We tested whether a dense residual-stream direction that reads human versus machine authorship also writes AI-like style. In Qwen2.5-7B base and instruct checkpoints, the selected direction achieves 0.9997 held-out AUROC. Adding it during generation, however, produces small, uncertain, and detector-inconsistent changes; nuisance residualization, matched controls, and a reduced base replication do not alter that conclusion. Endpoint quality measures show no broad collapse that would trivially mask the effect.

The key takeaway is that linear readability is not evidence of causal writability. A powered follow-up should use 64–100 held-out prompts per condition, select causal layers on a separate generation-validation set, intervene across adjacent layers, compare addition with projection removal, and obtain blinded human or current-model judgments. Replication in a second model family and directions fit from the target model’s own outputs would separate cross-generator authorship structure from write-side alignment. Sparse-feature decomposition is most informative after establishing that a dense causal effect exists.

References

- Christopher Ackerman and Nina Panickssery. Inspection and control of self-generated-text recognition ability in Llama3-8B-Instruct. In *International Conference on Learning Representations*, 2025.
- Diaoulé Diallo, Katharina Dworatzky, Sophie Jentzsch, Peer Schütt, Sabine Theis, and Tobias Hecking. The effectiveness of style vectors for steering large language models: A human evaluation, 2026.
- Liam Dugan, Alyssa Hwang, Filip Trhlik, Josh Magnus Ludan, Andrew Zhu, Hainiu Xu, Daphne Ippolito, and Chris Callison-Burch. RAID: A shared benchmark for robust evaluation of machine-generated text detectors. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie, Yuxuan Ding, Jianwei Yue, and Yupeng Wu. How close is ChatGPT to human experts? comparison corpus, evaluation, and detection, 2023.
- Kristian Kuznetsov, Laida Kushnareva, Polina Druzhinina, Anton Razzhigaev, Anastasia Voznyuk, Irina Piontkovskaya, Evgeny Burnaev, and Serguei Barannikov. Feature-level insights into artificial text detection with sparse autoencoders. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025.
- Yafu Li, Qintong Li, Leyang Cui, Wei Bi, Zhilin Wang, Longyue Wang, Linyi Yang, Shuming Shi, and Yue Zhang. MAGE: Machine-generated text detection in the wild. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The assistant axis: Situating and stabilizing the default persona of language models, 2026.

OpenAI. GPT-2 output detector model. <https://huggingface.co/openai-community/roberta-base-openai-detector>, 2019. Model card, revision 6cba99c003b711c7fe94f8a3aa2be35a792cb6fa.

Gerrit Quaremba, Hanqi Yan, Elizabeth Black, Denny Vrandečić, and Elena Simperl. Linear probing provides robust and efficient detection of machine-generated text, 2026.

Qwen Team. Qwen2.5 Technical Report, 2024.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.

Alex Reinhardt, Ben Markey, Michael Laudenbach, Kachata Pantusen, Ronald Yurko, Gordon Weinberg, and David West Brown. Do LLMs write like humans? variation in grammatical and rhetorical styles. *Proceedings of the National Academy of Sciences*, 2025. doi: 10.1073/pnas.2422455122.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering, 2023.

Mikhail Vishnyakov and Tatiana Gaintseva. SV-Detect: Ai-generated text detection with steering vectors, 2026.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency, 2023.

A Reproducibility Details

All stages use seed 42. The instruct generation sweep takes 376.9 seconds after direction construction, and the reduced base sweep takes 75.0 seconds. Expensive stages checkpoint intermediate artifacts. Model weights remain local and are not committed. From the repository root, the analysis is reproduced with:

```
python src/eda.py
python src/experiment.py \
  --model artifacts/models/Qwen2.5-7B-Instruct \
  --tag qwen25_7b_instruct --instruct --full-conditions \
  --n-train 240 --n-val 80 --n-test 80 --n-prompts 16 \
  --batch-size 4 --max-new-tokens 80
python src/experiment.py \
  --model artifacts/models/Qwen2.5-7B \
  --tag qwen25_7b_base \
  --n-train 240 --n-val 80 --n-test 80 --n-prompts 12 \
  --batch-size 4 --max-new-tokens 80
python src/evaluate.py \
  --tags qwen25_7b_instruct qwen25_7b_base --skip-judge
python src/add_perplexity.py
python -m src.analyze
```

The final validation audit confirms the required files, split sizes, lack of document leakage, generation row counts, prompt counts, complete metrics, valid score ranges and confidence intervals, residual-vector orthogonality, and matched vector norms. Raw and derived outputs are stored under `results/`, figures under `figures/`, and execution logs under `logs/`.

B Additional Results

B.1 Detector validation

Both detectors exceed chance on held-out MAGE and no-attack RAID samples (table 4), but neither is a perfect measure of perceived authorship. The public detector’s 0.623 RAID accuracy, despite 0.868 AUROC, also illustrates that a fixed threshold can be poorly calibrated under distribution shift.

Table 4: Independent detector validation before causal interpretation. The no-attack RAID sample contains 400 texts; the lexical detector uses a larger MAGE test sample than the neural detector for computational efficiency.

Detector	Validation set	n	AUROC	Accuracy
MAGE TF-IDF	MAGE test	12,000	0.792	0.711
MAGE TF-IDF	RAID no attack	400	0.906	0.835
Public RoBERTa	MAGE test	1,000	0.739	0.673
Public RoBERTa	RAID no attack	400	0.868	0.623

B.2 Readout across layers

Table 5: Held-out AUROC by decoder layer. Validation AUROC selects layer 5 in both checkpoints even though later layers have slightly higher test AUROC; test data were not used for selection.

Checkpoint	Decoder layer				
	5	9	14	18	22
QWEN2.5-7B-INSTRUCT	0.9997	1.0000	1.0000	1.0000	1.0000
QWEN2.5-7B	0.9997	1.0000	1.0000	1.0000	1.0000

Table 5 reports test AUROC across the prespecified layer grid. These values are descriptive after validation-only selection and do not justify switching the causal intervention to a test-favored layer.

B.3 Prompt-only baseline

Table 6: Corrected prompt-only “human style” baseline relative to unsteered instruct generation ($n = 16$). The instruction was placed before the long prefix so truncation could not remove it. The detectors disagree, while similarity falls and repetition rises.

Measure	Prompt minus unsteered	95% CI	Sign-flip p
MAGE $P(\text{AI})$	-0.022	$[-0.133, +0.091]$	.724
Public RoBERTa $P(\text{AI})$	+0.060	$[-0.096, +0.213]$	.490
Word count	+4.88	$[-9.44, +18.63]$	.531
Repeated 3-gram rate	+0.212	$[+0.064, +0.380]$	.021
Semantic similarity	-0.505	$[-0.624, -0.392]$	< .001
Base-Qwen perplexity	+1817.23	$[-12.36, +5469.01]$	.984

The first prompt-baseline implementation placed its instruction after a long prefix, and truncation removed the instruction. Validation caught this issue. We regenerated all 16 rows with the instruction first; table 6 contains only the corrected results. The extreme mean perplexity comes from several repetitive or pathological outputs and is correspondingly unstable.